— *Solution notes* —

COMPUTER SCIENCE TRIPOS Part IB – 2020 – Paper 5

6 Computer Networking (awm22)

Routing, NAT, addressing, WiFi subtopic in Topic 3 (data-link)

(*a*) Arriving at your college room, you plug into the wired Ethernet jack for the first time. The network admin has a record of your MAC address and your machine can join the network without further action on your part.

Assume: Your laptop's Ethernet address is 0a:0b:0c:0d:0e:0f, DHCP server address is 131.111.7.3, your IPv4 address will be 131.111.7.121, the gateway's IP address is 131.111.7.1, and Ethernet address is 00:01:02:03:04:05, the network netmask is 255.255.255.0

Write the series of protocol/packet exchanges that occur on the wired Ethernet link, up until you can send a single packet to 128.232.0.20. You do not need to describe packets after this packet has left the link. Include ARP and DHCP packets, stating the IP and Ethernet addresses of the packets where possible.

[10 marks]

Answer:

Essentially one mark per line; the extra information (MAC of router but IP of destination) of the last line and the correct ordering earns complete marks.

- Your laptop broadcasts a DHCP discover message from 0a:0b:0c:0d:0e:0f
- The DHCP server sends a DHCP offer of 131.111.7.121 to 0a:0b:0c:0d:0e:0f
- Client broadcasts a DHCP request for 131.111.7.121 from 0a:0b:0c:0d:0e:0f
- The DHCP server sends a DHCP ack to 0a:0b:0c:0d:0e:0f
- Your laptop sends an ARP request from 131.111.7.121/0a:0b:0c:0d:0e:0f for 131.111.7.1/ff:ff:ff:ff:ff:ff
- The gateway responds with an ARP reply from 131.111.7.1/00:01:02:03:04:05 to 131.111.7.121/00:11:22:33:44
- The laptop sends a packet from 131.111.7.121/0a:0b:0c:0d:0e:0f to 128.232.0.20 with Ethernet destination address 00:01:02:03:04:05

(*b*) Consider two neighbours, Alice and Bob. Each have wireless IPv4 routers with integrated NAT. Each neighbour connects their laptop to their own wireless router, and each uses appropriate utilities to examine the IP address of each laptop. They realise the laptops have the same IP address.

(*i*) How is that possible? [2 marks]

Answer: local addresses are entirely independent, allocated by each NAT box.

(*ii*) Justify one reason that wide-spread deployment of IPv6 would remove the need for the NAT devices.

[2 marks]

Answer: theoretically any device in either home could have its own unique IPv6 address enabling full Internet Routability

(*iii*) Justify one reason that an IPv6 user might want to continue using their NAT.

[2 marks]

Answer: NAT provide a firewall function; every device / port inside a NAT is no longer externally addressable without an explicit IGP (UPnP) type operation.

(*iv*) Further investigations show the two wireless routers are using the same wireless channel, although with different SSIDs. Detail what this situation means, why this situation is both possible and perfectly standards-compliant behaviour, and what implications there are for this situation — including how any negative effects can be made less severe.

[4 marks]

Answer: If the two access points are in close proximity, the use of the same channel will mean the two networks - despite being different SSIDs will be sharing the same Wifi communications channel.

It is perfectly standards compliant (2.4GHz is open use frequency) but the performance and security may be compromised.

Performance: because two AP's are competing in the same 802.11 channel.

security: because (as is always the case) a malicious neighbour could listen in to the network and if neighbour A can cause neighbour B's clients to join the wrong AP this opens up a further security risk too.